



Title

kdbwselect — Bandwidth Selection Procedures for Kernel Density Estimation and Inference.

Syntax

```
kdbwselect varname [if] [in] [, eval(gridvar) neval(#) kernel(kernelfn)
bwselect(bwmethod) bwcheck(#) imsegrid(#) separator(#) ]
```

Description

kdbwselect implements bandwidth selectors for kernel density point estimators and inference procedures developed in [Calonico, Cattaneo and Farrell \(2018\)](#). See also [Calonico, Cattaneo and Farrell \(2022\)](#) for related optimality results. It also implements other bandwidth selectors available in the literature.

A detailed introduction to this command is given in [Calonico, Cattaneo and Farrell \(2019\)](#).

Companion command is: [kdrobust](#) for kernel density point estimation and inference procedures.

Related software useful for empirical analysis is described in the following website:

<https://nppackages.github.io/>

Requires Stata 14 or later.

Options

eval(gridvar) specifies the grid of evaluation points for *xvar*. By default it uses 30 quantile-spaced points (deciles 0.1 through 0.9 in equal steps) over the support of *xvar*.

neval(#) specifies the number of quantile-spaced evaluation points used to estimate the density. Default is 30 evaluation points.

kernel(kernelfn) specifies the kernel function used. Options are: **epanechnikov**, and **uniform**. Default is **kernel**(epanechnikov).

bwselect(bwmethod) bandwidth selection procedure to be used. Options are:
mse-dpi second-generation DPI implementation of MSE-optimal bandwidth. Default choice.

imse-dpi second-generation DPI implementation of IMSE-optimal bandwidth.

imse-rot ROT implementation of IMSE-optimal bandwidth.

ce-dpi second generation DPI implementation of CE-optimal bandwidth.

ce-rot ROT implementation of CE-optimal bandwidth.

Note: MSE = Mean Square Error; IMSE = Integrated Mean Squared Error; CE = Coverage Error; DPI = Direct Plug-in; ROT = Rule-of-Thumb.

Default is **bwselect**(mse-dpi). For details on implementation see [Calonico, Cattaneo and Farrell \(2019\)](#).

bwcheck(#) specifies an optional positive integer so that the selected bandwidth is enlarged to have at least # effective observations available at each evaluation point.

imsegrid(#) number of evaluations points used to compute the IMSE bandwidth selector. Default is 30 points.

separator(#) draws separator line after every # variables; default is separator(5).

Example: Cholesterol Trial Data

```
Setup
. use nprobust_data.dta

MSE bandwidth selection procedure
. kdbwselect chol1 if t==0
```

Stored results

kdbwselect stores the following in **e()**:

Scalars

e(N) sample size after listwise deletion

Macros

e(cmd) command name
e(xvar) name of independent variable
e(bwselect) bandwidth selection choice
e(kernel) kernel choice

Matrices

e(bws) estimation result
e(eval) evaluation grid

References

- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2018. On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference. *Journal of the American Statistical Association*, 113(522): 767-779.
- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2019. nprobust: Nonparametric Kernel-Based Estimation and Robust Bias-Corrected Inference. *Journal of Statistical Software*, 91(8): 1-33. doi: 10.18637/jss.v091.i08.
- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2022. Coverage Error Optimal Confidence Intervals for Local Polynomial Regression. *Bernoulli*, 28(4): 2998-3022.

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